

Supplementary material to the RESOLVE Swiss statistical analysis plan: derivation of the power formula

The statistical analysis plan evaluates power with

$$\text{power} = 1 - T_{\nu,\lambda}(t_{1-\alpha/2,\nu}) + T_{\nu,\lambda}(-t_{1-\alpha/2,\nu}), \quad \lambda = \frac{\delta}{\sqrt{\sigma^2(1-r^2) \text{DE} \left(\frac{1}{\bar{m}k_1} + \frac{1}{\bar{m}k_2} \right)}}, \quad (1)$$

where δ is the target difference, σ the between-participant standard deviation of the outcome, r the baseline-to-follow-up correlation, DE the design effect, $\bar{m}$ the mean number of analysed participants per practice, k_1 and k_2 the numbers of practices per arm, $\nu = k_1 + k_2 - 2$, and $T_{\nu,\lambda}$ the distribution function of the noncentral t distribution with ν degrees of freedom and noncentrality λ . This note derives the formula in five elementary steps. Each step is standard, and the formula is their composition.

Step 1: the estimator and its variance

The planning test compares the two arm means, $\bar{y}_1 - \bar{y}_0$. In one arm with k practices of average size $\bar{m}$, clustering inflates the variance of the arm mean by the design effect,

$$\text{Var}(\bar{y}) = \frac{\sigma^2 \text{DE}}{\bar{m}k}, \quad \text{DE} = 1 + \left[\left(\text{cv}^2 \frac{k-1}{k} + 1 \right) \bar{m} - 1 \right] \rho, \quad (2)$$

with ρ the intra-cluster correlation and cv the coefficient of variation of practice size [1, eq. 1]. The two arms are independent, so their variances add:

$$\text{Var}(\bar{y}_1 - \bar{y}_0) = \sigma^2 \text{DE} \left(\frac{1}{\bar{m}k_1} + \frac{1}{\bar{m}k_2} \right). \quad (3)$$

This is the variance behind the sample size formula of Hayes and Moulton [2, eq. 7.12], with the per-arm terms added separately for unequal numbers of clusters as in their section 7.6.2.

Step 2: the baseline

The analysis uses the baseline measurement of the outcome, which multiplies the relevant residual variance by the factor $1 - r^2$ [3]. The true standard error of the comparison is therefore

$$\text{SE} = \sqrt{\sigma^2(1-r^2) \text{DE} \left(\frac{1}{\bar{m}k_1} + \frac{1}{\bar{m}k_2} \right)}, \quad (4)$$

the expression under the root in the noncentrality parameter of eq. (1).

Step 3: if the standard error were known

With known SE the statistic $T = (\bar{y}_1 - \bar{y}_0)/\text{SE}$ would be standard normal under the null hypothesis and normal with mean $\lambda = \delta/\text{SE}$ and unit variance under a true difference of δ . Power would be $\Phi(\lambda - z_{1-\alpha/2})$. This is the normal approximation, and it is the basis of the requirement table in the analysis plan.

Step 4: the standard error is estimated from few practices

In a cluster randomised trial the scarce ingredient of $\widehat{\text{SE}}$ is the between-practice variance. It is estimated from the $k_1 + k_2$ practice-level summaries, and their pooled variance carries $\nu = k_1 + k_2 - 2$ degrees of freedom [2, section 10.3.1], so that $s^2 \sim \sigma_{\text{SE}}^2 \chi_\nu^2 / \nu$ independently of the numerator. A ratio of the form

$$T = \frac{Z + \lambda}{\sqrt{\chi_\nu^2 / \nu}}, \quad Z \sim N(0, 1) \text{ independent of } \chi_\nu^2, \quad (5)$$

is by definition noncentral t with ν degrees of freedom and noncentrality λ . The test statistic has exactly this structure: the numerator is the estimate, which in units of SE is a standard normal shifted by λ , and the denominator is the estimated standard error divided by the true one. Under the null hypothesis $\lambda = 0$, the distribution is central t_ν , and the test rejects when $|T| > t_{1-\alpha/2, \nu}$.

Step 5: power is the rejection probability under λ

$$\text{power} = P(|T| > t_{1-\alpha/2, \nu}) = 1 - T_{\nu, \lambda}(t_{1-\alpha/2, \nu}) + T_{\nu, \lambda}(-t_{1-\alpha/2, \nu}), \quad (6)$$

which is eq. (1). This last step is the two-sample power computation of Chow et al. [4, section 3.2.1], applied at the practice level: their degrees of freedom $n_1 + n_2 - 2$ for individuals become $k_1 + k_2 - 2$ for practices, and their standard error becomes eq. (4).

Assumptions

Three approximations are inherited and are declared in the analysis plan. The practice-level summaries are treated as normally distributed with a common variance, which is the assumption of the underlying t comparison [2, section 10.3.1]. The design effect with unequal practice sizes is the approximation of Eldridge et al. [1, eq. 1], exact for the size-weighted comparison of practice means and conservative for individual-level analyses. The baseline factor $1 - r^2$ is applied at the individual level, where a formula carrying the individual-level and the cluster-level baseline correlations separately exists [5]. Evaluating power on the t scale rather than the normal scale builds the small-sample correction into the planning, as current guidance recommends [6, 7].

The script `R/sample_size.R` in the repository implements eq. (1) line by line, and the simulations in `R/simulations/` confirm it empirically.

References

- [1] Sandra M Eldridge, Deborah Ashby, and Sally Kerry. Sample size for cluster randomized trials: effect of coefficient of variation of cluster size and analysis method. *International Journal of Epidemiology*, 35(5):1292–1300, 2006. doi: 10.1093/ije/dyl129.
- [2] Richard J. Hayes and Lawrence H. Moulton. *Cluster Randomised Trials*. Chapman and Hall/CRC, Boca Raton, 2 edition, 2017. doi: 10.4324/9781315370286.
- [3] George F. Borm, Jaap Fransen, and Wim A. J. G. Lemmens. A simple sample size formula for analysis of covariance in randomized clinical trials. *Journal of Clinical Epidemiology*, 60(12):1234–1238, 2007. doi: 10.1016/j.jclinepi.2007.02.006.
- [4] Shein-Chung Chow, Jun Shao, Hansheng Wang, and Yuliya Lokhnygina, editors. *Sample Size Calculations in Clinical Research*. Chapman & Hall/CRC, Boca Raton, 3rd edition, 2018.

- [5] Steven Teerenstra, Sandra Eldridge, Maud Graff, Esther de Hoop, and George F. Borm. A simple sample size formula for analysis of covariance in cluster randomized trials. *Statistics in Medicine*, 31(20):2169–2178, 2012. doi: 10.1002/sim.5352.
- [6] Clémence Leyrat, Katy E Morgan, Baptiste Leurent, and Brennan C Kahan. Cluster randomized trials with a small number of clusters: which analyses should be used? *International Journal of Epidemiology*, 47(1):321–331, 2018. doi: 10.1093/ije/dyx169.
- [7] Gerard J P van Breukelen and Math J J M Candel. How to design and analyse cluster randomized trials with a small number of clusters? comment on leyrat et al. *International Journal of Epidemiology*, 47(3):998–1001, 2018. doi: 10.1093/ije/dyy061.